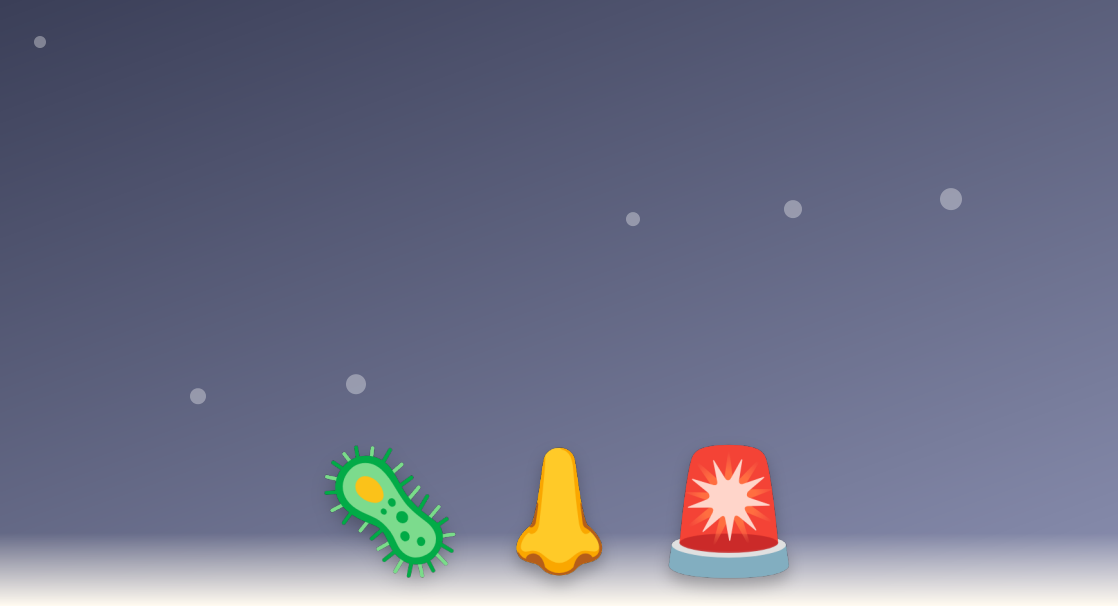


Defenders of the Body

II PUC · Ch 7 · Human Health and Disease

Starring Captain WBC 🦸 & the Immune Squad

A BAIO storybook by Narendra Sir · Biology Lecturer

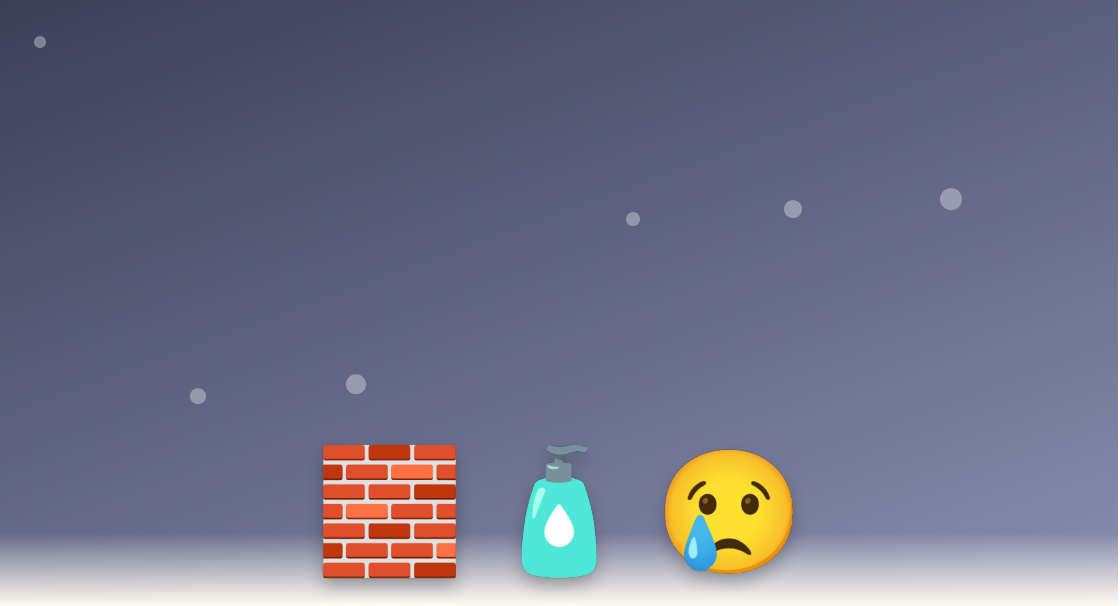


An Intruder Sneaks In

A tiny virus slipped into the body through the nose. Alarm signals spread through the bloodstream. "Intruder alert!" shouted Captain WBC, leader of the Immune Squad. "Any organism that causes disease is a **pathogen** — bacteria, viruses, fungi, protozoans or worms. Our job is **immunity**: the body's power to fight them."

💡 **REMEMBER**

Common cold is caused by rhinoviruses; it infects the nose and respiratory passage but not the lungs.



PAGE 2 OF 8

The Walls and Gates

The first defence was always ready — **innate immunity**, present from birth and non-specific. The skin was a **physical barrier**, and **mucus** in the nose trapped the virus. Acid in the stomach, saliva and tears were **physiological barriers**. "Not everyone gets stopped at the gate," the Captain admitted, "so the guards come next."

💡 REMEMBER

Innate immunity has four barriers: physical, physiological, cellular and cytokine barriers.



The Guards Attack

Guards rushed in: **neutrophils** (PMNL), **monocytes** and **natural killer cells**, with **macrophages** swallowing invaders whole — the **cellular barriers**. Virus-infected cells sent out warning proteins called **interferons**, which protect nearby healthy cells — the **cytokine barrier**. The area grew warm and swollen — inflammation, the sign of a battle.

💡 **REMEMBER**

Interferons are proteins secreted by virus-infected cells that protect non-infected cells from further infection.



The Special Forces

The virus kept multiplying, so the Captain called the special forces — **acquired immunity**, which is specific and has **memory**. Two teams were trained: **B-lymphocytes** and **T-lymphocytes**. "Our first meeting with this virus gives a slow **primary response**," said the Captain. "Next time it will be fast and strong — the **secondary response**, also called the anamnestic response."

💡 **REMEMBER**

B and T lymphocytes are produced in the bone marrow; T cells mature in the thymus.



Antibodies Lock On

B-cells released Y-shaped proteins called **antibodies**, each made of two heavy and two light chains — H_2L_2. Each antibody recognised one specific **antigen** on the virus and locked onto it. This is the **humoral immune response**, as antibodies travel in the blood. Meanwhile T-cells attacked infected cells directly — the **cell-mediated immunity** that also causes graft rejection.

💡 **REMEMBER**

Types of antibodies: IgA, IgM, IgE, IgG and IgD.



Active and Passive

"When the body makes its own antibodies, it is **active immunity**," said the Captain. "When ready-made antibodies are given, it is **passive immunity** — like the **colostrum** in a mother's first milk, rich in IgA, or the antivenom given after a snakebite." **Vaccination** gives weakened or inactivated germs so the body makes **memory cells** without falling ill.

💡 **REMEMBER**

Recombinant DNA technology produces vaccines in bacteria or yeast, e.g. the hepatitis B vaccine from yeast.



When Defenders Misfire

Sometimes the squad overreacts. Harmless pollen or dust can trigger an **allergy**: **IgE** antibodies make **mast cells** release **histamine** and serotonin, causing sneezing and watery eyes. Sometimes the squad even attacks the body's own cells — **autoimmunity**, as in rheumatoid arthritis. The squad trains in **primary lymphoid organs** (bone marrow, thymus) and gathers in **secondary lymphoid organs** like the spleen and lymph nodes.

💡 **REMEMBER**

MALT (mucosa-associated lymphoid tissue) makes up about 50 percent of the lymphoid tissue in the human body.



Memory Cells Remember

The virus was finally defeated. But a few **memory cells** stayed behind, holding the virus's 'photograph'. Months later the same virus returned — and was destroyed within hours, before any sneezing began. "That is why vaccines are so useful," said Captain WBC. "They train us before the real battle."

💡 **REMEMBER**

Immunisation programmes have eradicated smallpox and nearly eliminated polio.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

pathogen p.1 · immunity p.1 · innate immunity p.2 · physical barrier p.2 · mucus p.2 · physiological barriers p.2 · neutrophils p.3 · monocytes p.3 · natural killer cells p.3 · macrophages p.3 · cellular barriers p.3 · interferons p.3 · cytokine barrier p.3 · acquired immunity p.4 · memory p.4 · B-lymphocytes p.4 · T-lymphocytes p.4 · primary response p.4 · secondary response p.4 · antibodies p.5 · antigen p.5 · humoral immune response p.5 · cell-mediated immunity p.5 · active immunity p.6 · passive immunity p.6 · colostrum p.6 · Vaccination p.6 · memory cells p.6 · allergy p.7 · IgE p.7 · mast cells p.7 · histamine p.7 · autoimmunity p.7 · primary lymphoid organs p.7 · secondary lymphoid organs p.7



QUICK QUIZ

Do you remember?

1. Which cells help fight infection?

- Red blood cells
- White blood cells
- Platelets
- Nerve cells

2. What is the role of antibodies?

- To carry oxygen
- To recognise and bind specific antigens
- To digest food
- To clot blood

3. Why are vaccines useful?

- They kill all bacteria
- They produce memory cells for faster protection
- They replace white blood cells
- They cure allergies

4. Antibodies are produced by:

- T-lymphocytes
- B-lymphocytes
- Neutrophils
- Red blood cells

5. Colostrum gives the baby:

- Active immunity
- Passive immunity
- Autoimmunity
- No immunity

Answers: 1-White blood cells · 2-To recognise and bind specific antigens · 3-They produce memory cells for faster protection · 4-B-lymphocytes · 5-Passive immunity